Coursework Specification

Read this coursework specification carefully, it tells you how you are going to be assessed, how to submit your coursework on-time and how (and when) you'll receive your marks and feedback.

Module Code	CSI-7-MAL
Module Title	Machine Learning
Lecturer	Enrico Grisan
% of Module Mark	100 %
Distributed	[25/10/23]
Submission Method	Submit online via this Module's Moodle site
Submission Deadline	[13/12/23 23:59]
Release of Feedback & Marks	Feedback and provisional marks will be available in the Gradebook on Moodle

Coursework Aim:

The final coursework is a **group work,** requiring the students to work on the assigned machine learning problem, that they need to tackle from setting up the data to the evaluation and discussion of the performance of the proposed solution.

The aim of the coursework is to show the ability of the student to set up a machine learning processing pipeline, starting from the raw data and ending with a critical appraisal of the results obtained, especially in the light of other works applied on the same or similar set of data, and to present their work thoroughly and professionally.
The work should therefore present all the steps required to prepare and process the data, with a reasoned description of all the relevant choices performed when alternative routes were available.

The expected submission will contain:

1) a journal-style report describing in detail the problem and the proposed solution

2) The jupyter notebook, and/or Pyhon modules and scripts used to solve the problem and generate the figures/graphs used in the report

3) an additional appraisal of the contributions of the members of the group will be requested, and the final group mark will be adjusted considering each member's contribution.

Coursework Details:

Type:	Journal paper styled report.
Group dimension	3-5 students
Word Count:	Limit is 8 pages using the provided IEEE template and formatting. The page limit is including figures, charts and bibliography but excluding code that shall be provided as appendix or as a separate Python files (no screenshots in the report). As a guide, 8 pages are roughly equivalent to 4000-4500 words. All text exceeding 8 pages will not be considered for the marking.
Presentation:	▪ The report must be submitted as a Word document (.doc/docx) or a PDF ▪ Coursework must be submitted according to the provided template ▪ The report must be referenced, and a bibliography provided ▪ The report needs to contain at least: - an introductory section with a general presentation of the problem with relevant background - a literature review presenting the state of the art and related works - a method section containing a detailed description of all preprocessing and processing steps and validation/test strategy - a result section presenting the results obtained and the performance of the ML pipeline developed/applied - a discussion section containing a critical appraisal of the proposed solution ▪ Your student number and name must appear as author information .
Referencing:	IEEE Style referencing should be used
Regulations:	Make sure you understand the University Regulations on expected academic practice and academic misconduct. Note in particular: ▪ Your work must be your own. Markers will be attentive to both the plausibility of the sources provided as well as the consistency and approach to writing of the work. Simply, if you

	do the research and reading, and then write it up on your own, giving the reference to sources, you will approach the work in the appropriate way and will cause not give markers reason to question the authenticity of the work. ▪ All quotations must be credited and properly referenced. Paraphrasing is still regarded as plagiarism if you fail to acknowledge the source for the ideas being expressed. **Use of Generative AI tools** According to the University Regulations, you take responsibility that any submitted work is your own. Copying and pasting text from a generative AI tool is tantamount to plagiarism, while using it for drafting or revising your work is an acceptable use. Moreover double check the content of the generated text, as often times is incorrect or patently wrong or referencing to non existent facts or sources. In case we spot this, you will be deducted marks for not reporting facts correctly or wrongly claiming to have read sources AND you will be subject to an academic misconduct investigation. **TURNITIN:** When you upload your work to the Moodle site it will be checked by anti-plagiarism software.

Learning Outcomes

This coursework will fully assess the following learning outcomes for this module.

Knowledge and understanding:
-demonstrating a systematic understanding of the domain of machine learning algorithms including the importance of research, methodologies, driving innovation and contribution;
- producing and reviewing research informed work which applies and is at the forefront of the developments in the domain

Intellectual skills:
- critically analysing a case-based domain using appropriate analytic and quantitative methods;
- developing the in-depth knowledge necessary to identify and apply suitable techniques in order to synthesize advanced theory/practical concepts

Practical skills:
- evaluating a project applying appropriate techniques conducting effective independent research
- developing the in-depth knowledge necessary to identify machine learning project domains and apply suitable techniques in order to synthesize advanced (theory/practical) concepts to design, develop, deploy, and maintain bespoke/innovative machine learning solutions using suitable tools e.g: Python;

Transferable Skills:
- evaluating existing/emerging machine learning technology and techniques, carrying out independent research, recognizing and contributing to opportunities for innovation, synthesise ideas/theories/solutions and report back appropriately to your peers
- self-managing study time and work effectively to meet deadlines;
- selecting and evaluate supporting resources/tools for a particular purpose, as well as being able to make effective contributions as team member/leader when required.

Assessment Criteria and Weighting

LSBU marking criteria have been developed to help tutors give you clear and helpful feedback on your work. They will be applied to your work to help you understand what you have accomplished, how any mark given was arrived at, and how you can improve your work in future.

	Criteria	Feedforward comments 100-80%	79-70%	69-60%	59-50%	49-40%	39-30%	29-0%
	1. Research Systematic identification and use of academic and relevant resources	Extensive independent relevant research evidenced by quality and quantity used. Ability to draw on own research and that of others.	Extensive independent relevant research evidenced by quality and quantity used. Some autonomous research.	Wide range of relevant sources identified and used. Very little guidance needed.	A range of sources identified and used. Limited guidance needed.	Limited research identified and used. Some guidance needed to complete research tasks.	Some evidence of research but insufficient amount. Needs support to develop research skills.	Little or no research presented. Needs significant support to develop research skills.
	2. Subject Knowledge Understanding and application of subject knowledge. Contribution to subject debate.	Shows sustained breadth, accuracy and detail in understanding key aspects of subject. Contributes to subject debate. Awareness of ambiguities and limitations of knowledge.	Shows breadth, accuracy and detail in understanding key aspects of subject. Contributes to subject debate. Some awareness of ambiguities and limitations of knowledge.	Accurate and extensive understanding of key aspects of subject. Evidence of coherent knowledge.	Accurate understanding of key aspects of subject. Evidence of coherent knowledge.	Understanding of key aspects of subject. Some evidence of coherent knowledge.	Some evidence of superficial understanding of subject. Inaccuracies.	Little or no evidence of understanding of subject. Inaccuracies.
	3. Critical Analysis Analysis and interpretation of sources, literature and/or results. Structuring of issues/debates.	Very high-quality analysis developed independently. Sustained evaluation and synthesis of resources. Use of evidence-based arguments. Thoroughly identifies trends, inconsistency, congruence, and states the implications.	Sustained evaluation and synthesis of resources. Use of evidence-based arguments. Thoroughly identifies trends, inconsistency, congruence, and states the implications.	Evaluation and synthesis of resources. Use of evidence-based arguments. Identifies trends, inconsistency, congruence, and states the implications.	Evaluation and synthesis of resources. Use of evidence-based arguments.	Some attempt at evaluation and synthesis of resources. Some use of evidence-based arguments.	Limited evaluation of resources. Limited use of evidence-based arguments	Little or no evaluation of resources. Very little use of evidence-based arguments.
	4. Testing and Problem-Solving Skills Design, implementation, testing and analysis of **product/process/system/ idea/solution(s)** to practical or theoretical questions or problems	Autonomous creation and novel implementation. Adapts to unforeseen practical and theoretical challenges to achieve identified goals.	Almost entirely autonomous creation and implementation. Adapts to unforeseen practical and theoretical challenges to achieve identified goals.	Mainly autonomous creation and implementation. Adapts to unforeseen practical and theoretical challenges to achieve identified goals.	Some autonomy to create and implement. Some adaption made to unforeseen practical and theoretical challenges to achieve identified goals.	Exploration of possible solution(s). Use of established approaches to resolve practical and theoretical problems.	Limited exploration of possible solution(s) using established approaches to resolve practical and theoretical problems.	Little or no exploration of solution(s). Question or problem unresolved.
	5. Experimentation Problem-solving and exploration of the assessment including risk taking and testing of ideas.	Evidence of rich, in-depth exploration and highly innovative problem-solving. Responds to assessment in a way that pushes boundaries and/or makes exciting new connections possible within discipline. Finds a balance between experimentation and resolution to create a highly coherent and innovative piece of work.	Evidence of in-depth exploration and innovative problem-solving. Responds to assessment in a way that pushes boundaries and/or makes new connections possible within discipline. Finds a balance between experimentation and resolution to create a coherent and innovative piece of work.	Clear evidence of exploration and experimentation. The work may not find the balance between experimentation and the attempt to create a coherent piece of work.	There is some problem-solving evident but lacks creative exploration and experimentation.	Some evidence of limited problem-solving. Appropriate response to assessment but there is little development and exploration evident.	Some evidence of engagement with assessment but it is insufficiently developed. Unable to engage with the challenges of assessment.	Little or no evidence of engagement with the assessment, it shows little development.
	6. Practical Competence Skills to apply theory to practice or to test theory	Consistently applies comprehensive practical and/or technical skills which reflect the correct application of theory to context.	Applies comprehensive practical and/or technical skills which reflect the correct application of theory to context.	Applies practical and/or technical skills very well and with careful application of theory to context.	Applies practical and/or technical skills correctly and with some application of theory to context with only minimal errors.	Applies a sufficient level of practical and/or technical skills but with some errors. Limited application of theory to context.	Applies some aspects of the practical and/or technical skills but these are incomplete or contain important errors. Very limited application of theory to context.	Limited application of some aspects of the practical and/or technical skills. No application of theory to context.

	7. Academic Integrity[1] Acknowledges and gives credit to the work of others follows the conventions and practices of the discipline including appropriate use of referencing standards for discipline.	Consistent, error free application of relevant referencing conventions with great attention to detail.	Consistent, error free application of relevant referencing conventions.	Consistent application of relevant referencing conventions with few errors.	Application of relevant referencing conventions, with some errors and / or inconsistencies.	Generally correct application of relevant referencing conventions, with some errors and / or inconsistencies.	Limited application of referencing conventions and / or errors.	Very limited or no application of referencing conventions, and/or multiple errors.

[1] **The application of this criterion is independent of the process outlined in the Student Academic Misconduct Procedure**

How to get help

We will discuss this Coursework Specification in class. However, if you have related questions, please contact me (Enrico Grisan: enrico.grisan@lsbu.ac.uk as soon as possible.

Quality assurance of coursework specifications

Coursework specifications within CSI division go through internal (for new modules with 100% coursework also through external) moderation. This is to ensure high quality, consistency and appropriateness of the coursework as well as to share best practice within the CSI division.

Details of the moderators for this coursework specification are below:

Moderated (internal)	[Name, date]
Moderated (CSI lead)	[Name, date]
Signed off by (HoD/DHoD)	[Name, date]